① a) $f(x) = \operatorname{sen} x$

$$f(x)' = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \Rightarrow \lim_{\Delta x \to 0} \frac{\operatorname{sen}(x+\Delta x) - \operatorname{sen} x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\operatorname{sen} x . \cos \Delta x + \operatorname{sen} \Delta x . \cos x - \operatorname{sen} x}{\Delta x} = \text{(I)}$$

$$\text{(I)} \lim_{\Delta x \to 0} \frac{\operatorname{sen} x . \overbrace{(\cos \Delta x - 1)}^{0}}{\cancel{\Delta x}} + \frac{\overbrace{\cancel{\operatorname{sen} \Delta x}}^{1} . \cos x}{\cancel{\Delta x}} = \lim_{\Delta x \to 0} \cos x \Rightarrow f(x)' = \cos x$$

b) $f(x) = 3x^2 \to f(x)' = 6x$ // c) $f(x) = \ln x \to f(x)' = \frac{1}{x}$ d) $f(x) = e^x$ $f(x)' = e^x$

*Verificar demonstração.

② a) $f(x) = \cos x$ $f(x)' = \lim_{\Delta x \to 0} \frac{\cos(x+\Delta x) - \cos x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\cos x . \cos \Delta x - \operatorname{sen} x . \operatorname{sen} \Delta x - \cos x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\cos x . \overbrace{(\cos \Delta x - 1)}^{0}}{\cancel{\Delta x}} - \frac{\operatorname{sen} x . \overbrace{\cancel{\operatorname{sen} \Delta x}}^{1}}{\cancel{\Delta x}} = -\operatorname{sen} x \Rightarrow f(x)' = -\operatorname{sen} x$

$x_0 = \frac{\pi}{6}$ $f(x)' = -\frac{1}{2}$

b) $f(x) = \operatorname{tg}(x)$ $\operatorname{tg}(x) = \frac{\operatorname{sen}(x)}{\cos(x)}$ *Agora, podemos usar a regra do quociente para a derivada: $x_0 = \frac{\pi}{4} \to \frac{d}{dx}[\operatorname{tg} x] = 2$

$$\text{(I)} \frac{d}{dx}\left[\frac{\operatorname{sen} x}{\cos x}\right] = \frac{(\operatorname{sen} x)' . \cos x - \operatorname{sen} x (\cos x)'}{\cos^2 x} = \frac{\cos^2 x + \operatorname{sen}^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} \Rightarrow \frac{d}{dx}[\operatorname{tg} x] = \frac{1}{\cos^2 x} \text{ //}$$

c) $$f(x) = \sqrt{x} = \frac{d}{dx}[\sqrt{x}] = \lim_{\Delta x \to 0} \frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x} \cdot \frac{\sqrt{x+\Delta x} + \sqrt{x}}{\sqrt{x+\Delta x} + \sqrt{x}} = \frac{\cancel{\Delta x}}{\cancel{\Delta x}(\sqrt{x+\Delta x} + \sqrt{x})} = \lim_{\Delta x \to 0} \frac{1}{\sqrt{x+\Delta x} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \cdot \frac{\sqrt{x}}{\sqrt{x}} = \frac{\sqrt{x}}{2x} \text{ //} \Rightarrow \frac{d}{dx}[\sqrt{x}] = \frac{\sqrt{x}}{2x}$$

$x_0 = 1 \to f(x)' = \frac{1}{2}$ //

d) $f(x) = 2^x \to f(x)' = x . 2^{(x-1)}$ $x_0 = 1 \to f(x)' = 1$ //

e) $f(x) = e^x \to f(x)' = e^x$ $x_0 = 2 \to e^2$ //

③ a) $s(t) = t^2 + t - 2$

$t_0 = 2$

Derivando

VELOCIDADE: $\frac{ds}{dt} = 2t + 1$ e em $t_0 = 2$, temos uma velocidade de $5 m/s$

Aceleração: $\frac{dv}{dt} (2t+1) = \frac{dv}{dt} 2$, ou seja, a aceleração é contínua em $2 m/s^2$

b) $s(t) = \frac{1}{t}$, $t_0 = 3$

VELOCIDADE: $\frac{ds}{dt}\left(\frac{1}{t}\right) = -t^{-2} = -\frac{1}{9} m/s$

ACELERAÇÃO: $\frac{dv}{dt}\left[-1 . t^{-2}\right] = (-1).(-2). t^{-3} = 2t^{-3} = \frac{2}{27} m/s^2$

c) $s(t) = \sqrt[3]{t}$, $t_0 = 2$

VELOCIDADE: $\frac{ds}{dt}\left(t^{\frac{1}{3}}\right) = \left[\frac{1}{3} . t^{\frac{-2}{3}}\right] = \frac{1}{3} \cdot \frac{1}{t^{\frac{2}{3}}} = \frac{1}{3 . \sqrt[3]{4}}$

ACELERAÇÃO: $\frac{dv}{dt}\left[\frac{1}{3} . t^{\frac{-2}{3}}\right] = -\frac{2}{9} . t^{\frac{-5}{3}} = -\frac{2}{9} \cdot \frac{1}{\sqrt[3]{t^5}} = -\frac{2}{9} \cdot \frac{1}{\sqrt[3]{32}} = \frac{-2}{9 . \sqrt[3]{32}} m/s^2$

④

a) $f(x) = x^2 - 2x$, $x_0 = 1$ $\quad y = 1^2 - 2x = -1$

$f(x)' = 2x - 2 = m \to m = 2.1 - 2 \Rightarrow m = 0$

$y - y_0 = m.(x - x_0) \to y + 1 = 0.(x - 1) \to \boxed{y = -1}$

b) $f(x) = \operatorname{sen} x$, $x_0 = 0$ $\quad y = \operatorname{sen} 0 \to y = 0$

$m = f(x)' \to m = \cos x$, no ponto $x_0 = 0 \to m = 1$

$y - y_0 = m.(x - x_0) \to y = 1.(x) \to \boxed{y = x}$

c) $f(x) = \sqrt{x}$, $x_0 = 4$ $\quad y = 2$

$m = \frac{1}{2\sqrt{x}}$ no ponto desejado: $m = \frac{1}{4}$

$y - 2 = \frac{1}{4}.(x - 4) \to \boxed{y = \frac{x+4}{4}}$

d) $f(x) = \frac{1}{x}$, $x_0 = 1$ $\quad y = 1$

$m = x^{-1} = -1.x^{-2} = \frac{-1}{x^2} \to m = -1$

$y - 1 = -1.(x - 1) \to \boxed{y = -x + 2}$

⑤ *A derivada da soma, é a soma das derivadas:

a) $f(x) = x + 1 \to f(x)' = 1$

b) $f(x) = x^2 + 3 \to f(x)' = 2x$

c) $f(x) = \operatorname{sen} x + \cos x \to f(x)' = \cos - \operatorname{sen} x$

d) $f(x) = 8x^{11} - \frac{7}{3x^3} - \frac{\sqrt{3}}{7} \to f(x)' = 88x^{10} + 7x^{-4}$

e) $f(x) = 3 + 2x^m + x^{2m} \to f(x)' = 2m\,x^{(m-1)} + 2m\,x^{(2m-1)}$

f) $f(x) = x^2 - 4x^{-11} - \ln x \to f(x)' = 2x + 44x^{-12} - \frac{1}{x}$

⑥ $f'(x) = u'(x).v(x) + u(x).v'(x)$

a) $f(x) = (x^2+1).(x^3+2x) \to f(x)' = x.(x^3+2x) + (x^2+1).(3x^2+2) \to f(x)' = 4x^4 + 5x^2 + 2x + 2$

b) $f(x) = \operatorname{sen} x . \cos x \to f'(x) = \cos x . \cos x - \operatorname{sen} x . \operatorname{sen} x \to f'(x) = \cos^2 x - \operatorname{sen}^2 x$

c) $f(x) = x^2 . \operatorname{sen} x \to f'(x) = 2x + \cos x$

d) $f(x) = e^{2x} \to f'(x) = 2e^{2x}$

e) $f(x) = e^x . \cos x \to f'(x) = e^x . \cos x - \operatorname{sen} . e^x = e^x . (\cos x - \operatorname{sen} x)$

f) $f(x) = e^x . \operatorname{sen} x \to f'(x) = e^x . \cos + e^x . \operatorname{sen} x = e^x . (\cos x + \operatorname{sen} x)$

g) $f(x) = x^3 . \ln(x) \to f'(x) = 3x^2 . \ln x + x^3 . \frac{1}{x} = 3x^2 \ln x + x^2 = x^2 . (3\ln x + 1)$

⑦ $f(x) = x^3 . e^{-x}$ no ponto $x_0 = 0$

$m = 3x^2 . e^{-x} + e^{-x} . x^3$ e $x = 0$ $\therefore$ $m = 0$

$\boxed{y = 0}$ ↳ Reta tangente.

⑧ REGRA DO QUOCIENTE: $\frac{u'(x).v(x) - v'(x).u(x)}{[v(x)]^2}$

a) $f(x) = \frac{e^x}{x^2} \to f'(x) = \frac{e^x.x^2 - 2x.e^x}{x^4} = \frac{e^x.(x^2 - 2x)}{x^4}$

b) $f(x) = \frac{x^2+1}{x+1} \to f'(x) = \frac{2x.(x+1) - (x^2+1).1}{(x+1)^2} = \frac{x^2+2x-1}{(x+1)^2}$

c) $f(x) = \frac{x+1}{x-1} \to f'(x) = \frac{1.(x-1) - 1.(x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2}$

d) $f(x) = \frac{\operatorname{sen} x}{a^x} \to f'(x) = \frac{\cos x . a^x - a^x . \ln a . \operatorname{sen} x}{a^{2x}} = \frac{\cos x - \ln a\, \operatorname{sen} x}{a^x}$

e) $f(x) = tg x = \frac{sen x}{cos x} \to f'(x) = \frac{cos . cos + sen . sen}{cos^2} = \frac{\overset{sec(x)}{1}}{cos^2}$

f) $f(x) = \frac{1}{sen x} \to f'(x) = \frac{-cos}{sen^2 x}$

g) $f(x) = \frac{1}{[u(x)]^n} \to f'(x) = \frac{0 . u(x)^n - n . u(x)^{(n-1)} . 0}{[u(x)]^{2n}} = 0$

h) $f(x) = \frac{x^2 . sen x}{e^x} \to f'(x) = \frac{2x . sen x + cos x . x^2}{e^x} = \frac{u(x)}{v(x)} =$

$= \frac{u'(x) . v(x) - u(x) . v'(x)}{[v(x)]^2}$

i) $f(x) = \frac{\overset{\to u(x)}{cos x}}{\underset{\to v(x)}{x . e^x}} \to f'(x) = \frac{cos x}{e^x + x . e^x}$

$f'(x) = \frac{-sen x . x e^x - cos x . (e^x (1+x))}{x^2 . e^{2x}}$

j) $f(x) = \frac{1}{cos x} \to f'(x) = \frac{sen x}{cos^2 x} = tg(x) . sec(x)$

k) $f(x) = \frac{tg x}{sen x + cos x} \to f'(x) = \frac{tg(x)}{sen x + cos x} = \frac{sec^2 . (sen + cos x) - (cos x - sen x) . tg}{(sen x + cos x)^2}$

l) $f(x) = \frac{x+3}{x-1} + \frac{(x+2)}{x+1} = \frac{-1}{(x+1)^2} \to f'(x) = \frac{0 . (x+1)^2 - 2 . (x+1)}{(x+1)^4}$ = * CONTINUAR

9) $f(x) = \frac{1}{x} + e^x$ em $x_0 = -1 \to y = -1 + e^{-1}$

$m = -\frac{1}{x^2} + e^x \to x_0 = -1 \therefore m = -1 + e^{-1}$ $\therefore$ $\boxed{y = (-1+e^{-1}) . (x+1) + (-1+e^{-1})}$

10) $y = \frac{x^2}{x^2+1}$

a) $\frac{2x . (x^2+1) - x^2 . (2x)}{(x^2+1)^2} = \frac{2x^3 + 2x - 2x^3}{(x^2+1)^2} = \frac{2x}{(x^2+1)^2}$

b) $\lim_{x \to \infty} \frac{x^2}{x^2+1} \to \lim_{x \to \infty} \frac{\cancel{x^2}}{\cancel{x^2} (1+\frac{1}{x^2})} \to \lim_{x \to \infty} \frac{1}{1} = 1$

c) Passa pela origem no ponto $(0;0) \to m = \frac{2x}{(x^2+1)^2}$ $\boxed{x=0, \ y=0}$

A reta tg $\to y - y_0 = m . (x - x_0) \to \frac{x^2}{x^2+1} = \frac{2x}{(x^2+1)^2} . x =$

$= x^2 . (x^2+1) = 2x^2 \to x^2 + 1 = 2 \to x = \pm 1$ *Agora substitui no $f(x)$.

$\boxed{x=0, \ y=0}$ $\boxed{x=1, \ y=\frac{1}{2}}$ $\boxed{x=-1, \ y=\frac{1}{2}}$ → Esses são os pontos que a tg passa pela origem

11) Regra da cadeia: $F(x) = g(f(x)) \Rightarrow F'(x) = g'(f(x)) . f'(x)$

a) $f(x) = cos 2x \to f'(x) = -sen(2x) . 2$

b) $f(x) = sen^3 x \to f'(x) = 3 . sen^2(x) . cos x$

c) $f(x) = cos^n x \to f'(x) = n . cos^{n-1} x . (-sen x)$
↳ temos $u = cos x$ e $v = u^n$

d) $f(x) = cos x^n \to f'(x) = -sen(x^n) . n . x^{n-1}$
↳ $cos x$ e $u = x^n$

e) $f(x) = ln(x^2+1) \to f'(x) = \frac{2x}{x^2+1}$

f) $f(x) = a^{cos x} \to f'(x) = -sen(x) . ln a . a^{cos x}$
↳ $\frac{d}{dx}(a^{u(x)}) = a^{u(x)} . ln a . u'(x)$

g) $f(x) = e^{7x^2-2x} \to f'(x) = e^{7x^2-2x} . (14x-2)$

h) $f(x) = cos(e^x) \to f'(x) = -sen(e^x) . e^x$

i) $f(x) = e^{cos 2x} \to f'(x) = e^{cos 2x} . (-sen(2x) . 2)$

j) $f(x) = \sqrt{sen x} \to f'(x) = \frac{cos x}{2\sqrt{sen x}}$

12) $s = \sqrt{4t^2+3}$ $t > 0$

$\frac{ds}{dt} = \frac{8t}{2\sqrt{4t^2+3}} = \frac{4t}{\sqrt{4t^2+3}} \to v(t) = \frac{4t}{\sqrt{4t^2+3}}$

*A derivada nos deu exatamente a equação para conseguir calcular a velocidade instantânea.

a) $v = 0 \to t = 0$

b) $\frac{4t}{\sqrt{4t^2+3}} = 1 \to 4t = \sqrt{4t^2+3}$

$16t^2 = 4t^2 + 3 \to t^2 = \frac{\cancel{3}^{1}}{\cancel{12}_{4}} \to \boxed{t = \frac{1}{2}}$

c) $\frac{4t}{\sqrt{4t^2+3}} = 2 \to 4t = 2 . (\sqrt{4t^2+3}) \to 0 = 3$

*Isso quer dizer que não há tempo em que velocidade = 2.

(13) $s=\sqrt{5+t^2}$ $\quad \frac{ds}{dt} = \frac{t}{\sqrt{5+t^2}} \to v(t) = \frac{t}{\sqrt{5+t^2}}$

a) $\frac{t}{\sqrt{5+t^2}} = 0 \to t=0$ $\quad$ b) $\frac{t}{\sqrt{5+t^2}} = 1 \to t^2 = 5+t^2 \to 0=5$ $\quad$ c) $\frac{t}{\sqrt{5+t^2}} = 2 \to t^2 = 4.(5+t^2) \to t^2 = 20+4t^2 \to$

* não existe

$\to t^2 = -\frac{20}{3}$ $\quad$ * Impossível.

* A função é sempre positiva e menor que 1.

(14) $C=6+4\sqrt{x}$

a) $\frac{d}{dx}\left[6+4\sqrt{x}\right] \to \frac{4}{2\sqrt{x}} = \frac{2}{\sqrt{x}} = \frac{2}{\sqrt{16}} = \frac{1}{2}$ $\quad$ b) $\frac{2}{\sqrt{x}} = 0{,}4 \to 20 = 4\sqrt{x} \to 400 = 16x \to x = 25$

(15) $\frac{d}{dx}\left[40+3x+9\sqrt{2x}\right] \to 3+\frac{9}{\sqrt{2x}}$

a) $3+\frac{9}{2.10} \to 3+\frac{9}{10} \to \boxed{3{,}9}$ $\quad$ b) $4{,}5 = 3+\frac{9}{\sqrt{2x}} \to \sqrt{2x} = \frac{90}{15} \to 2x=36 \to \boxed{x=18}$

(16) a) $V_m = \frac{\Delta s}{\Delta t} \to V_m = \frac{s(9)-s(0)}{t(9)-t(0)} = \frac{1152}{9} = 128\,m/s$

b) $\frac{ds}{dt}\left[200t - 5t^2 - \frac{t^3}{3}\right] = \boxed{200-10t-t^2}$ e quando $t=0$ $v=200\,m/s$ $\quad$ c) $t=5s$ $v=125\,m/s$

d) $\text{ACELERAÇÃO} = \frac{\text{VELOCIDADE}}{\text{TEMPO}}$ $\quad a = \frac{s'(9)-s'(0)}{9} = \frac{29-200}{9} = -19\,m/s^2$ $\quad$ e) $f''(x) = -10-2t \to -22\,m/s^2$

(17) $f(x) = \frac{e^x+e^{-x}}{2}$ $\quad x=-2$ $\; y = \frac{e^{-2}+e^{2}}{2}$ $\quad f'(x) = \frac{(e^x-e^{-x})}{2} = m$ $\quad \boxed{y = \left(\frac{e^{-2}+e^{2}}{2}\right) + \left|\frac{(e^x-e^{-x})}{2}\right|.(x+2)}$

$\frac{\Delta x}{x} = u$ $\quad \Delta x = u.x$

(19) $\frac{d}{dx}\ln x \to \lim_{\Delta x \to 0} \frac{\ln(x+\Delta x)-\ln(x)}{\Delta x} = \frac{\ln\left(\frac{x+\Delta x}{x}\right)}{\Delta x} = \frac{\ln\left(1+\frac{\Delta x}{x}\right)}{\Delta x} = \frac{\overset{\to 1}{\ln(1+u)}}{x\,u} = \frac{1}{x}$

(20) $\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$

a) $f(x) = \text{arcsen}\,x \to$ $\quad x = \text{sen}\,w$

$\frac{d}{dx}\text{arcsen}\,x = \frac{1}{\cos w}$ $\to$ Sendo o inverso de arcsen x : sen

* Agora escrevendo em função de x

$\frac{d}{dx}\text{arcsen}\,x = \frac{1}{\sqrt{1-\text{sen}^2 w}} \to \boxed{\frac{d}{dx}\text{arcsen}\,x = \frac{1}{\sqrt{1-x^2}}}$

b) $f(x) = \arccos x \to \cos z = x$ $\quad$ * Esse é o inverso.

$\frac{d}{dx}\arccos x = \frac{1}{-\text{sen}\,z} \to \frac{d}{dx}\arccos x = \frac{1}{-\sqrt{1-\cos^2 z}} = \frac{1}{-\sqrt{1-x^2}}$

DERIVADA DO INVERSA

c) $f(x) = \text{arctg}\,x$ $\quad \boxed{x = \text{tg}\,\beta}$ $\to$ Esse é o inverso.

$\frac{d}{dx}\text{arctg}\,x = \frac{1}{\frac{1}{\cos^2\beta}} \to \frac{d}{dx}\text{arctg}\,x = \frac{1}{\sec^2\beta} \to \frac{d}{dx}\text{arctg}\,x = \frac{1}{1+x^2}$

$\boxed{1+\text{tg}^2\beta}$

* Para a realidade do exercício, tem

$\cos^2 + \text{sen}^2 = 1 \to \frac{\cos^2}{\cos^2} + \frac{\text{sen}^2}{\cos^2} = \frac{1}{\cos^2} \to \boxed{\sec^2 = 1+\text{tg}^2}$

(21) $\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$ $\quad \frac{dy}{dx}$ → Qual a taxa de variação de y quando x se altera.

a) $f(x) = \arcsen(3x)$ $\quad \sen y = 3x$

$$\frac{dy}{dx}\sen y = \frac{d}{dx}3x \to \frac{dy}{dx}\cdot \cos y = 3 \to$$

$$\to \frac{dy}{dx} = \frac{3}{\cos y} \to \frac{dy}{dx} = \frac{3}{\sqrt{1-\sen^2 y}} \to \boxed{\frac{dy}{dx} = \frac{3}{\sqrt{1-9x^2}}}$$

b) $f(x) = \arccos(x^2)$ $\quad \cos y = x^2$

$$\frac{dy}{dx}(\cos y)' = \frac{d}{dx}(x^2) \to -\sen \frac{dy}{dx} = 2x \to$$

$$\to \frac{dy}{dx} = -\frac{2x}{\sen y} \to \frac{dy}{dx} = -\frac{2x}{\sqrt{1-\cos^2 y}} \therefore \boxed{\frac{dy}{dx} = -\frac{2x}{\sqrt{1-x^4}}}$$

c) $f(x) = \arctg\frac{1}{x}$ $\quad \boxed{tg\, y = \frac{1}{x}}$ → Equação original $\quad \boxed{\sec^2 y = 1 + tg^2 y}$

$$\frac{d}{dx} tg\, y = \frac{d}{dx}\frac{1}{x} \to \sec^2 y \cdot \frac{dy}{dx} = -\frac{1}{x^2} \to \frac{dy}{dx} = -\frac{1}{x^2 \cdot \sec^2 y} \to \frac{dy}{dx} = \frac{1}{x^2(1+tg^2 y)} \to \frac{dy}{dx} = \frac{1}{\cancel{x^2}\cdot\left(\frac{x^2+1}{\cancel{x^2}}\right)} \therefore \boxed{\frac{dy}{dx} = \frac{1}{x^2+1}}$$

(22) $\boxed{[\ln(u)]' = \frac{1}{u}\cdot u'}$

a) $f(x) = \ln|-3x|$

$f'(x) = \frac{1}{-3x}\cdot -3$; $f'(x) = x^{-1}$ $\quad \therefore f(x) = (-1).x^{!}.x^{n-1}$

$f''(x) = -1x^{-2}$ $\quad f'''(x) = 2x^{-3}$ $\quad$ É um fatorial.

$f^{IV} = -6x^{-4}$ $\quad f^{V} = 24x^{-5}$

b) $f(x) = \frac{5}{e^{2x-7}} \Rightarrow f(x) = 5.e^{-2x+7}$

$f'(x) = 5.e^{-2x+7}.(-2) = -10e^{-2x+7}$

$f''(x) = e^{-2x+7}.(-10).(-2) = 20e^{-2x+7}$ $\quad \therefore f^{(n)}(x) = (-2)^n.5e^{-2x+7}$

$f'''(x) = e^{-2x+7}.(20).(-2) = -40e^{-2x+7}$

(23) a) $\lim\limits_{x\to 0}\frac{x}{e^x-1} \to \lim\limits_{x\to 0}\frac{1}{e^x} = 1$

b) $\lim\limits_{x\to 2}\frac{x^2+x-6}{x^2-3x+2} \to \lim\limits_{x\to 2}\frac{2x+1}{2x-3} = 5$

c) $\lim\limits_{x\to 0}\frac{\sen x - x}{e^x+e^{-x}-2} \to \lim\limits_{x\to 0}\frac{\cos x-1}{e^x-e^{-x}} \to \lim\limits_{x\to 0}\frac{-\sen x}{e^x+e^x}$

$\to \lim\limits_{x\to 0}\frac{-\sen x}{e^x+e^x} = \frac{0}{2} = 0_{//}$

d) $\lim\limits_{x\to\infty}\frac{e^x-1}{x^3+4x} \to \lim\limits_{x\to\infty}\frac{e^x}{3x^2+4} = +\infty$

e) $\lim\limits_{x\to\infty} x.\sen\frac{1}{x} = 0$

f) $\lim\limits_{x\to 0}\left(\frac{1}{x^2+x} - \frac{1}{\cos x-1}\right) \to \lim\limits_{x\to 0}\frac{(\cos x-1)-(x^2+x)}{(x^2+x).(\cos x-1)} \to \lim\limits_{x\to 0}\frac{-\sen-(2x+1)}{(2x+1).(\cos x-1)+(x^2+x)(-\sen)} = \frac{-1}{0} = -\infty$

g) $\lim\limits_{x\to 0}\frac{x^2-\sen^2 x}{\sen^2 x} \to \frac{2x-2\sen x.\cos x}{2\sen x.\cos x} \to \lim\limits_{x\to 0}\frac{x-\sen x.\cos x}{\sen x.\cos x} \to \lim\limits_{x\to 0}\frac{1-(\cos^2 x-\sen^2 x)}{\cos^2 x-\sen^2 x} = \frac{1-1}{1} = 0$

(24) a) $W = \lim\limits_{x\to 0^+}\left(x^{\frac{1}{tg x}}\right) = 0$

*Aplicamos o log em ambos lados

$\ln W = \lim\limits_{x\to 0^+}\frac{\ln x}{tg x} = -\infty \Rightarrow e^{-\infty} = 0$

$tg\, 0^+$ → quase zero, mas não é zero.
$\ln 0^+ \to -\infty$, pois vai gerar expoente negativo para gerar um x próximo a zero.

*Pode usar log de qualquer base.

b) $N = \lim\limits_{x\to 1} x^{\frac{1}{1-x}} = \frac{1}{e}$

*Aplicando o log: L'HOSPITAL

$\ln N = \lim\limits_{x\to 1}\frac{\ln x}{1-x} \Rightarrow \ln N = \lim\limits_{x\to 1}\frac{1}{-x} = -1 \Rightarrow e^{-1} = \frac{1}{e}$

c) $P = \lim\limits_{x\to 0^+}(2x^2+x)^x = 1$

*Aplicando o log

$\ln P = \lim\limits_{x\to 0^+} x.\ln(2x^2+x) = 0 \Rightarrow e^0 = 1$

$$\left(\ln(u)\right)' = \frac{1}{u} \cdot u'$$

d) $L = \lim_{x\to\infty} \left(1+\frac{1}{2x}\right)^{x} = e^{\frac{1}{2}}$ //

$$\ln L = \lim_{x\to\infty} \frac{\ln\left(1+\frac{1}{2x}\right)}{\frac{1}{x}} \xrightarrow{\text{L'HOSPITAL}} \ln L = \lim_{x\to\infty} \frac{1}{2+\cancel{\frac{1}{x}}_{0}} = e^{\frac{1}{2}} \;//$$

(I) $\ln\left(1+\frac{1}{2x}\right)' = \left(\frac{1}{1+\frac{1}{2x}}\right)\cdot\left(-\frac{1}{2x^2}\right) = -\frac{1}{x\cdot(2x+1)}$

(II) $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$

(III) $\lim_{x\to 0} \frac{\frac{-1}{x\cdot(2x+1)}}{-\frac{1}{x^2}} = \frac{x^{\cancel{2}}}{\cancel{x}\cdot(2x+1)} =$

$$= \frac{x}{2x+1} = \frac{\cancel{x}}{\cancel{x}\cdot\left(2+\frac{1}{x}\right)} = \frac{1}{2+\frac{1}{x}}$$

(I) $[\ln(\cos 2x)]' = \frac{1}{\cos 2x} \cdot -2\,\text{sen}(2x) = -2\,tg(2x)$

e) $b = \lim_{x\to 0} (\cos 2x)^{\frac{3}{x^2}} =$

$$\ln b = \lim_{x\to 0} \frac{3\ln(\cos 2x)}{x^2} \xrightarrow{\text{L'HOSPITAL (I)}} \frac{3\cdot(-2\,tg(x))}{2x} = \frac{-6\,tg\,x}{2x} =$$

$$\ln b = \lim_{x\to 0} \frac{-3\,tg(2x)}{x} = \lim_{x\to 0} -3\cdot\sec^2(2x)\cdot 2 =$$

$$= \ln b = \lim_{x\to 0} \frac{-6}{\cancel{\cos^2 2x}_{1}} \Longrightarrow \lim_{x\to 0} (\cos 2x)^{\frac{3}{x^2}} = e^{-6}$$

(25) $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ a) $\frac{dy}{dx} = \frac{\cos t}{-\text{sen}\, t} = -\cot g$

b) $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2\cdot\cancel{\text{sen}\, t}\cdot\cos t}{3\cdot\cos^2 t\cdot\cancel{\text{sen}\, t}} = \frac{2}{3\cdot\cos t}$

(26) PONTO $(2;2)$

$t = \frac{2}{x}$ $\quad y = 3t^2 - 1$

$t = \frac{2}{2} = 1$ $\quad 2 = 3t^2 - 1 \to t = 1$

$$\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{dy}{dx} = \frac{6t}{-2t^{-2}} = -3t^{-3} \to m = -3$$

$t = 1$

$y - 2 = -3\cdot(x-2)$

$\boxed{y = -3x + 8}$

(27) * Derivar implicitamente para encontrar $\frac{dy}{dx}$, que é o coeficiente angular. Depois, calculamos as coordenadas desejadas.

$$x^2 + y^2 = 1 \Longrightarrow \frac{dx^2}{dx} + \frac{dy^2}{dx} = \frac{d1}{dx} \Longrightarrow 2x + 2y\frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = \frac{-x}{y} = m$$

$\boxed{y - y_0 = m\cdot(x - x_0)}$

$$c\left(\frac{\sqrt{3}}{2}; \frac{1}{2}\right) \quad \frac{dy}{dx} = \frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = -\sqrt{3} = m \Longrightarrow y - \frac{1}{2} = -\sqrt{3}\cdot\left(x - \frac{\sqrt{3}}{2}\right) \Longrightarrow \boxed{y = -\sqrt{3}\,x + 2}$$

(28) a) TAXA MÉDIA DE PRODUÇÃO:

$$\frac{P(3)-P(1)}{3-1} = \frac{5880}{2} = 2940$$

2940 MIL UNIDADES POR HORA

b) $\frac{dP}{dt}$ * Nessa questão, observamos que ele pede a razão de produção em um tempo específico e não o total fabricado nas 5 horas, sendo necessário derivar.

$\boxed{t=5}$ $\frac{dP}{dt} = 245\cdot(4t+4) \longrightarrow 245\cdot 24 = 5880$ mil/hora

$\boxed{t=4}$ $\frac{dP}{dt} = 183\cdot(2) \to 366$ mil/hora

c) $P(3) - P(2) = 7350 - 3920 = 3430$ mil peças

$P(7) - P(6) = 4392 - 4026 = 366$ mil peças.

(30) Mesmo lógica da questão anterior.

$$\frac{dx^3}{dx} + \frac{dy^3}{dx} = \frac{d9}{dx} \Rightarrow 2x^2 + 2y^2\frac{dy}{dx} = 0 \Rightarrow x^2 + y^2\frac{dy}{dx} = 0 \quad \therefore \frac{dy}{dx} = \frac{-x^2}{y^2} = m$$

$$y - 2 = -\frac{1}{4}\cdot(x-2) \Rightarrow \boxed{y = \frac{-x+9}{4}}$$

(31) $$4\cdot\frac{d}{dx}x^3y^5 + \frac{d}{dx}\cos 2y = \frac{d}{dx}tg(2x) + \frac{d}{dx}e^{3y} \Rightarrow 4\cdot\left(3x^2y^5 + 5x^3y^4\frac{dy}{dx}\right) - Sen(2y)\cdot 2\frac{dy}{dx} = 2\cdot sec^2(2x) + 3\cdot e^{3y}\cdot\frac{dy}{dx}$$

$$12x^2y^5 + 20x^3y^4\cdot\frac{dy}{dx} - Sen(2y)\cdot 2\frac{dy}{dx} - 3e^{3y}\cdot\frac{dy}{dx} = 2\,sec^2(2x)$$

$$\frac{dy}{dx}\cdot\left[20x^3y^4 - Sen(2y) - 3e^{3y}\right] = 2\,sec^2(2x)$$

$$\boxed{\frac{dy}{dx} = \frac{2\,Sec^2(2x)}{20x^3y^4 - Sen(2y) - 3e^{3y}}}$$

(33) $$\frac{d}{dx}[x\cos y] + \frac{d}{dx}[y\cdot\cos x] = \frac{d}{dx}[1] \Rightarrow \left(1\cdot\cos(y) - x\cdot Sen(y)\frac{dy}{dx}\right) + \left(\frac{dy}{dx}\cdot\cos(x) - y\,Sen(x)\right) = 0$$

$$\frac{dy}{dx}\left(-Sen(y) + \cos(x)\right) + \cos(y) - y\,Sen\,x = 0 \quad \therefore \boxed{\frac{dy}{dx} = \frac{y\,Sen\,x - \cos(y)}{\cos(x) - x\,Sen(y)}}$$

(34) a) $y = \pm\sqrt{4-x^2}$

b) $$\frac{dy}{dx} = \frac{d}{dx}\sqrt{4-x^2} \Rightarrow \frac{dy}{dx} = \frac{-x}{\sqrt{4-x^2}} \quad \Big| \quad \frac{dy}{dx} = \frac{d}{dx} - \sqrt{4-x^2} \Rightarrow \frac{dy}{dx} = \frac{x}{\sqrt{4-x^2}}$$

$$\frac{d}{dx}\sqrt{4-x^2} = \frac{1}{2\cdot\sqrt{4-x^2}}\cdot -2x = \frac{-x}{\sqrt{4-x^2}}$$

c) $$\frac{d}{dx}x^2 + \frac{d}{dx}y^2 = \frac{d}{dx}4 \Rightarrow 2x + 2y\frac{dy}{dx} = 0 \quad \therefore \boxed{\frac{dy}{dx} = \frac{-x}{y}}$$

d) $(1; +\sqrt{3})$ $\quad y - \sqrt{3} = -\sqrt{3}\cdot(x-1) \Rightarrow y = -\sqrt{3}x + 2\sqrt{3}$

$(1; -\sqrt{3})$ $\quad y + \sqrt{3} = \sqrt{3}\cdot(x-1) \Rightarrow y = \sqrt{3} - 2\sqrt{3}$

(35) Não, pois isso resulta em uma raiz negativa, ou seja, a curva nem existe.

(37) $16x^4 + y^4 = 32$

*DERIVANDO:

$$\frac{d}{dx}16x^4 + \frac{d}{dx}y^4 = \frac{d}{dx}32 \Longrightarrow 64x^3 + 4y^3\frac{dy}{dx} = 0 \Longrightarrow \frac{dy}{dx} = \frac{-64x^3}{4y^3} = \frac{-16x^3}{y^3}$$

No ponto $(1,2)$ $y - 2 = \frac{-16 \cdot 1^3}{2^3} \cdot (x-1) \Longrightarrow \boxed{y = -2x + 4}$

(38) $9x^3 - y^3 = 1 \Longrightarrow \frac{d}{dx}9x^3 - \frac{d}{dx}y^3 = \frac{d}{dx}1 \Longrightarrow 27x^2 - 3y^2\frac{dy}{dx} = 0 \Longrightarrow \frac{dy}{dx} = \frac{9x^2}{y^2}$

$m = \frac{9}{4}$ $\quad y - 2 = \frac{9}{2} \cdot (x-1) \Longrightarrow y = \frac{9}{2}(x-1) + 2$ reta tg

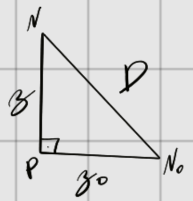
$m = -\frac{4}{9}$ $\quad \boxed{y = \frac{-4}{9}(x-1) + 2}$ —> Reta normal.

Temos que: $m_{normal} = -\frac{1}{m_{tg}}$

(39) $x^2 + xy + y^2 - 3y = 10 \Longrightarrow (x + 2y - 3)\frac{dy}{dx} = -(2x+y) \Longrightarrow \frac{dy}{dx} = \frac{-(2x+y)}{(x+2y-3)}$ no ponto $(2,3)$ $m_{tg}: -\frac{7}{5}$

$y - 3 = \frac{5}{7} \cdot (x-2) \Longrightarrow \boxed{y = \frac{5}{7}(x-2) + 3}$ $\quad \boxed{m_{normal}: \frac{5}{7}}$

(40) $z = 24t$ $\quad z_0 = 32 \cdot (t-2)$ * Note que temos os diferenças de tempo



$$D^2 = (z(t))^2 + (z_0(t))^2$$

$$\frac{d}{dx}D^2 = \frac{d}{dx}(z(t))^2 + (z_0(t))^2 \Longrightarrow 2D\frac{dD}{dt} = 2z(t)\frac{dz(t)}{dt} + 2z_0(t)\frac{dz_0(t)}{dt} \Longrightarrow \text{I}$$

$$\text{I} \quad \frac{dD}{dt} = \frac{z(t)\frac{dz(t)}{dt} + z_0(t)\frac{dz_0(t)}{dt}}{D}$$

↳ N_2 Parado

a) $t = 1$

$$\frac{dD}{dt}\ \frac{0.0 + 24.24}{24} = 24 \text{ m/s}$$

b) $t = 3$

$$\frac{dD}{dt}\ \frac{32.32 + 72.24}{\sqrt{6208}} = \frac{2752}{\sqrt{6208}} \text{ m/s}$$

(41) $3x^4 4y^2 - 7xy^3 = 4 - 8y$

$12x^4y^2 - 7xy^3 = 4 - 8y$

$\underbrace{\frac{d}{dx}12x^4y^2}_{\text{I}} - \underbrace{\frac{d}{dx}7xy^3}_{\text{II}} = \underbrace{\frac{d}{dx}4 - 8y}_{\text{III}}$

I $\quad 12 \cdot (4x^3 \cdot y^2 + 2yx^4) \to 48x^3y^2 + 24x^4y\frac{dy}{dx}$

II $\quad -7 \cdot (y^3 + 3y^2\frac{dy}{dx}) \to -7y^3 - 21xy^2\frac{dy}{dx}$

III $\quad 4 - 8y \to -8\frac{dy}{dx}$

$48x^3y^2 + 24x^4\frac{dy}{dx} - 7y^3 - 21y^2\frac{dy}{dx} = -8\frac{dy}{dx}$

$24x^4\frac{dy}{dx} - 21y^2\frac{dy}{dx} + 8\frac{dy}{dx} = -48x^3y^2 + 7y^3$

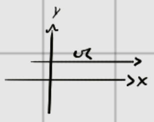
$$\boxed{\frac{dy}{dx} = \frac{7y^3 - 48x^3y^2}{24x^4y - 21y^2x + 8}}$$

(42) $7y^2 - xy^3 = 4 \xrightarrow{\text{Derivando}} \frac{d}{dx}7y^2 + \frac{d}{dx}xy^3 = \frac{d}{dx}4 \longrightarrow 14y\frac{dy}{dx} + \left(-y^3 + 3xy^2\frac{dy}{dx}\right) \longrightarrow 14\frac{dy}{dx} - y^3 - 3xy^2\frac{dy}{dx} = 0 \longrightarrow$ (*)

(*) $\frac{dy}{dx} \cdot (14 - 3xy^2) = y^3 \Longrightarrow \boxed{\frac{dy}{dx} = \frac{y^3}{14 - 3xy^2}}$

(43) r tg // eixo x

*Quando o coeficiente angular é zero



$x + (xy)^{\frac{1}{2}} + y = 1$ *DERIVANDO: $\frac{d}{dx}x + \underbrace{\frac{d}{dx}(xy)^{\frac{1}{2}}}_{(*)} + \frac{dy}{dx} = 1 =$ (I)

(*) $\frac{d}{dx}(xy)^{\frac{1}{2}} = \left(\frac{1}{2}(xy)^{-\frac{1}{2}}\right) \cdot \left(y + x\frac{dy}{dx}\right) =$

(I) $1 + \frac{dy}{dx} + \left(\frac{1}{2}(xy)^{-\frac{1}{2}}\right) \cdot \left(y + x\frac{dy}{dx}\right) + \frac{dy}{dx} = 0 \Longrightarrow \boxed{\frac{dy}{dx} = \frac{-2\sqrt{xy} - y}{x + 2\sqrt{xy}}}$

$\frac{dy}{dx} = 0$ quando $x.y = \frac{y^2}{4} \longrightarrow 4xy - y^2 = 0 \longrightarrow y.(4x - y) = 0$

$\boxed{y = 0 \text{ ou } y = 4x}$

Testando:

$y = 0 \Longrightarrow x + 0 + 0 = 1 \Rightarrow x = 1$

$y = 4x \Longrightarrow x + \sqrt{4x^2} + 4x = 1 \Rightarrow x = \frac{1}{7}$

PONTOS:

$(1; 0)$

$\left(\frac{1}{7}; \frac{4}{7}\right)$